Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/20/2006
PI Name:	Zhang, Nengli		
Project Title:	Simulation of Two-Phase Flow of Liquid Metal Using Nanofluids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
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Task Description:	To realize the simulation of liquid metal using nanofluids the most important key point is that the nanofluid used as the working fluid has high enough thermal conductivity. Copper-water nanofluids have the potential high thermal conductivity with suitable other thermophysical properties and, therefore, are selected the alternative liquids in the space experiments of the two-phase flow (TPF) of liquid metal. Since the lack of the thermophysical property data in literature, especially the thermal conductivity, for the nanofluids, the measurements of thermal conductivity of the nanofluids have to be performed first. Other properties, such as density, thermal capacity, and kinetic viscosity, can be calculated using available formulas for the time being. However, the wetting property of the nanofluids has to be measured for its important to the boiling performance of the fluids. The task includes the following studies: 1. Measurements of wetting characteristics, including spreading rate and evolution of contact angle. 2. Experimental study of the heat transfer performance of the nanofluids both in laminar and turbulent single flow region, Reynolds number		

	from 1,000 to 3,000. 3. Ground-based experimental study of the heat transfer performance of two-phase flow of the nanofluids. 4. Analysis of simulation of liquid metal using the nanofluids, including flow boiling, condensation, and phase separation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The research was interrupted for the change of NASA project. A feasibility study of simulation of two-phase flow of liquid metal using nanofluids have been conducted. The experimental measurements on low-concentration copper-water nanofluids have been made to varify the theoretical predictions for the nanofluids' thermal conductivity. The major study activities and results are summarized as follows. 1. An experimental setup: The new NanoFlash light flash system has been established which makes thermal properties, including thermal diffusivity, specific heat and thermal conductivity, testing fast easy and affordable both for solid and liquid samples. 2. The test runs of the measurement of thermal properties of low-concentration copper-water nanofluids (volume fraction of copper nanoparticles in water < 3 %) have been passed, which indicates that the NanoFlash light flash system can be successfully used to measure the thermal properties of nanofluids. 3. A data-base of thermal properties for relevant liquid metals (Li, K, Na) has been established. 4. The experimental work on identification of profiles near three-phase line of a sessile drop of 3% copper-water nanofluid by laser shadowgraphy method based on the catastrophe optical theory has been tested to demonstrate the method can successfully applied to measure the key parameters of nanofluids, such as the contact angle and the profile of three-phase line, which are important for the performance of two-phase flows. 5. The feasibility study of simulation of liquid metal flows using high-concentration nanofluids has been conducted and shows that in space Rankine-cycle power systems the single-phase liquid metal flows can be simulated using copper-water nanofluids, while the two-phase liquid-metal flow cannot be credibly simulated using surrogate fluids and the experimental tests in a subscale liquid-metal system are still indispensable.
Bibliography Type:	Description: (Last Updated:)